

DP-800: Developing AI-Enabled Database Solutions

From 12 March 2026 onwards

Design and develop database solutions (35–40%)	
Design and implement database objects	
1	Design and implement tables, including data types, size, columns, indexes, and column store indexes
2	Design and implement specialized tables, including in-memory, temporal, external, ledger, and graph
3	Design and implement JSON columns and indexes
4	Design and implement database constraints, including PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, and DEFAULT
5	Design and implement SEQUENCES
6	Design and implement partitioning for tables and indexes
Implement programmability objects	
7	Create views
8	Create scalar functions
9	Create table-valued functions
10	Create stored procedures
11	Create triggers
Write advanced T-SQL code	
12	Write common table expressions (CTEs)
13	Write queries that include window functions
14	Write queries that include JSON functions, such as JSON_OBJECT, JSON_ARRAY, JSON_ARRAYAGG, JSON_CONTAINS, OPENJSON, and JSON_VALUE
15	Write queries that include regular expressions, such as REGEXP_LIKE, REGEXP_REPLACE, REGEXP_SUBSTR, REGEXP_INSTR, REGEXP_COUNT, REGEXP_MATCHES, and REGEXP_SPLIT_TO_TABLE
16	Write queries that include fuzzy string matching functions, such as EDIT_DISTANCE, EDIT_DISTANCE_SIMILARITY, and JARO_WINKLER_DISTANCE
17	Write graph queries that use the MATCH operator
18	Write correlated queries
19	Implement error handling
Design and implement SQL solutions by using AI-assisted tools	
20	Interpret security impact of using AI-assisted tools
21	Enable GitHub Copilot and Microsoft Copilot in Fabric
22	Configure model and Model Context Protocol (MCP) tool options in a GitHub Copilot or Copilot in Fabric chat session
23	Create and configure GitHub Copilot instruction files
24	Connect to MCP server endpoints, including Microsoft SQL Server and Fabric lakehouse

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Secure, optimize, and deploy database solutions (35–40%)

Implement data security and compliance

- 25 Design and implement data encryption, including Always Encrypted and column-level encryption
- 26 Design and implement Dynamic Data Masking
- 27 Design and implement Row-Level Security (RLS)
- 28 Design and implement object-level permissions
- 29 Implement secure database access, including passwordless
- 30 Implement auditing
- 31 Secure model endpoints, including Managed Identity
- 32 Secure GraphQL, REST, and MCP endpoints

Optimize database performance

- 33 Recommend database configurations
- 34 Preserve data integrity and consistency by using transaction isolation levels and concurrency controls
- 35 Evaluate query performance by using query execution plans, dynamic management views (DMVs), Query Store, and Query Performance Insight
- 36 Identify and resolve query performance issues, including blocking and deadlocks

Implement CI/CD by using SQL Database Projects

- 37 Design and implement a testing strategy, including unit tests and integration tests
- 38 Create and manage reference/static data in source control
- 39 Create, build, and validate database models by using SQL Database Projects, including SDK-style models
- 40 Configure source control for SQL Database Projects
- 41 Manage branching, pull requests, and conflict resolution
- 42 Implement secrets management
- 43 Detect schema drift by using SQL Database Projects
- 44 Update an SQL database project and deploy changes
- 45 Design and implement controls for deployment pipelines, including branching policies, triggers in approvals, authentication tables, and code owners

Integrate SQL solutions with Azure services

- 46 Create configuration files for Data API builder (DAB)
- 47 Configure entities for REST and GraphQL, including data caching, pagination, searching, and filtering
- 48 Configure REST or GraphQL endpoints
- 49 Expose database objects, stored procedures, and views, including GraphQL relationships
- 50 Configure and implement DAB deployment
- 51 Recommend Azure Monitor configurations, including Application Insights and Log Analytics
- 52 Handle changes by using change event streaming (CES), change data capture (CDC), Change Tracking, Azure Functions with SQL trigger binding, or Azure Logic Apps

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Implement AI capabilities in database solutions (25–30%)

Design and implement models and embeddings

- 53 Evaluate external models, including multimodal, multilanguage, sizes, and structured output
- 54 Create and manage external models
- 55 Choose an embedding maintenance method, including table triggers, Change Tracking, Azure Functions with SQL trigger binding, Azure Logic Apps, CDC, CES, and Microsoft Foundry
- 56 Identify which columns to include in embeddings
- 57 Design and implement chunks for embeddings
- 58 Generate embeddings

Design and implement intelligent search

- 59 Choose from full-text, semantic vector, and hybrid search
- 60 Implement full-text search
- 61 Design for vector data, including vector data type, vector indexes, and size
- 62 Identify when to use vector-related types and functions for semantic searching, including VECTOR_NORMALIZE, VECTOR_DISTANCE, VECTORPROPERTY, and VECTOR_SEARCH
- 63 Choose between using ANN and ENN for vector search
- 64 Evaluate vector index types and metrics
- 65 Implement vector search
- 66 Implement hybrid search
- 67 Implement reciprocal rank fusion (RRF)
- 68 Evaluate performance of vector and hybrid search

Design and implement retrieval-augmented generation (RAG)

- 69 Identify use cases for RAG
- 70 Create a prompt by using the sp_invoke_external_rest_endpoint stored procedure
- 71 Convert structured data to JSON for language model processing
- 72 Send results to language model
- 73 Extract language model responses